

Can a jacket keep something cold?

Learning intention

Use evidence to explain thermal conduction and select insulation for a cold container.

A cold drink must arrive cool.

Your job: choose a jacket and defend it with evidence.

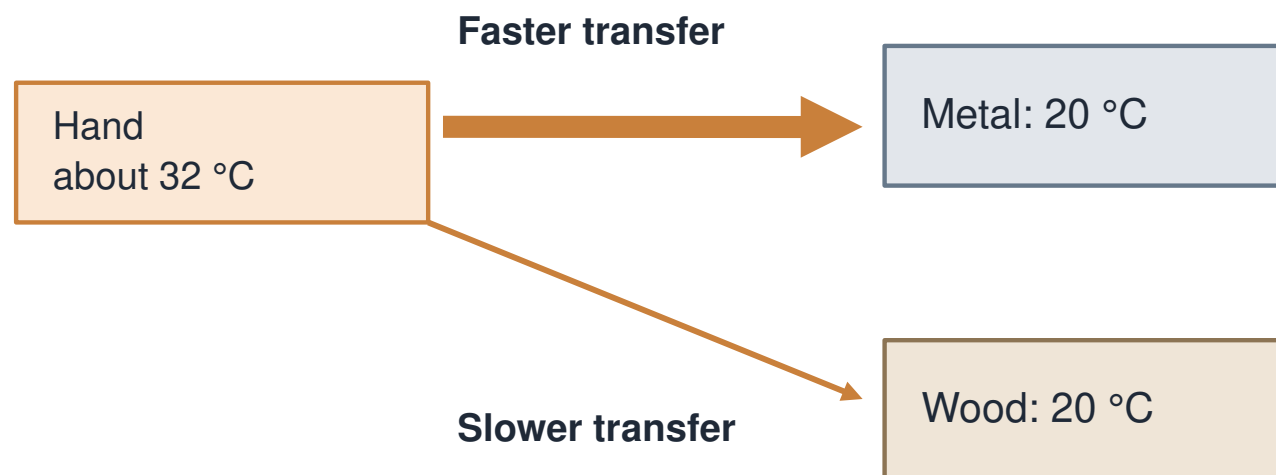
Today: notice → explain → test the claim.

What do we already know?

- R1. Name the material in a metal spoon.
- R2. Which is warmer: 8 °C or 20 °C?
- R3. Name a tool that measures temperature.

Feeling is not a thermometer

Both samples start at 20 °C

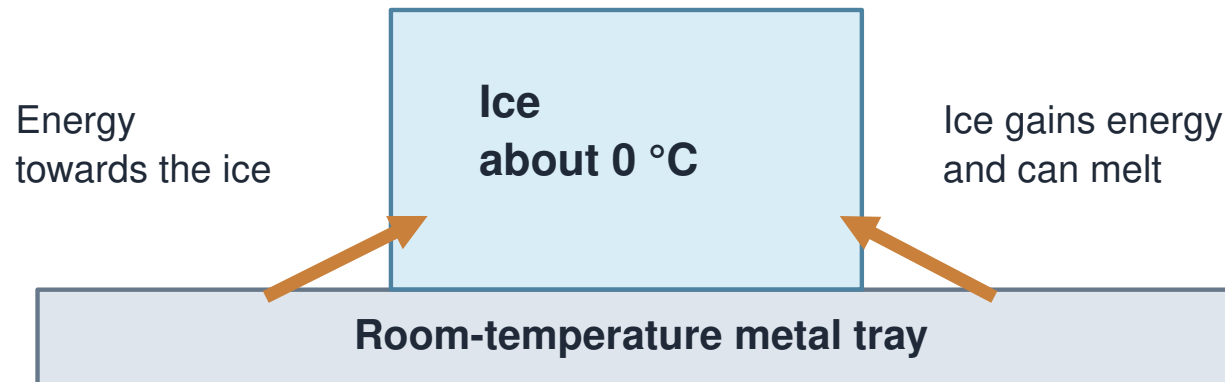


Arrow width is a model, not measured data.

The samples both start at 20 °C. A warmer hand loses energy to both. Metal transfers that energy more quickly.

Why can ice melt faster on metal?

Ice is cooler than the tray and room

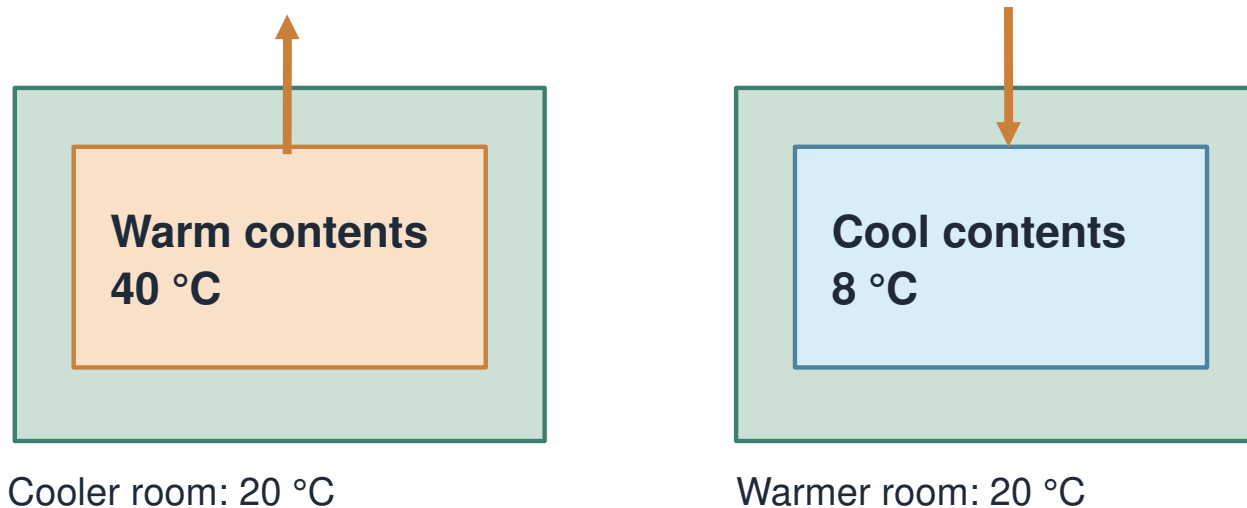


Model: other energy routes and sizes are simplified.

Energy transfers from warmer to cooler objects. A metal tray can transfer energy towards ice quickly. Melting does not mean the tray started hotter.

Insulation changes the rate

Insulation slows energy transfer



The green jacket adds insulation. It makes no heat.

Thermal insulators slow transfer by conduction. A jacket can slow cooling or warming. It does not create heat or stop transfer forever.

Evidence detectives: feeling versus measurement

D1. Both blocks measured 20 °C. Metal felt colder.

D2. In an ice test, more ice melted on metal.

Use page 1: reject, keep or improve each claim.

Which bottle warmed least?

Constructed sample data: identical sealed bottles. All start at 8 °C in a 20 °C room. Read the 20-minute temperature; justify one choice.

Jacket	Start / °C	20 min / °C
None	8	14
Cotton sleeve	8	12
Foam sleeve	8	10

Constructed sample data; equal bottles, fill, start temperature and room.

Improve the delivery claim

Claim: “Foam makes cold and is always best.”

Replace it with a sentence the evidence supports.

Name one extra check before a bigger decision.

A cold bottle wears an insulating jacket

The room is warmer than the bottle. What happens?

- A** The jacket produces cold.
- B** Energy still transfers in, but more slowly.
- C** The jacket stops all transfer forever.
- D** Energy transfers from cold bottle to warmer room.

Your design brief

Choose one response page: supported, standard or stretch.

Recommend a jacket using the common evidence.

Explain where energy moves and why.

Make the evidence do the work

Name the two objects at different temperatures.

Use at least one pair of values.

Finish: “This shows ... in this test.”

Check before you submit

Have I used “slows” rather than “stops”?

Does my arrow go from warmer to cooler?

Have I separated evidence from a guess?

Compare and improve

Show one sentence or arrow you improved.

Check a partner's evidence without judging their first idea.

What would make our conclusion stronger?

Our evidence has a boundary

We compared three particular jackets.

Thickness, fit and repeat results still matter.

Pupil choice: which jacket variable should be tested next?

Three things to leave with

- E1. Draw energy's direction: room 20 °C, bottle 8 °C.
- E2. Explain what a thermal insulator does.
- E3. Give one number that supports the foam choice.